

Wazuh SOC Lab

Security Operations Center (SOC) Project Report

Portfolio Project | Wazuh SIEM

Executive Summary

This project demonstrates the deployment and use of Wazuh as a Security Information and Event Management (SIEM) platform in a controlled virtual lab environment. The laboratory was designed to provide hands-on experience with endpoint monitoring, security event detection, alert investigation, malware and rootkit detection, and integration with external threat intelligence.

Project Objectives

- Deploy Wazuh in a controlled virtual environment.
- Configure a Wazuh server and a monitored endpoint.
- Collect and monitor security events from the endpoint.
- Investigate security alerts generated by simulated malicious activity.
- Use Suricata to provide network security monitoring.
- Integrate Wazuh with VirusTotal for additional threat intelligence context.
- Test detection capabilities using safe security-lab simulations.
- Develop practical SOC skills in alert monitoring and investigation.

Lab Architecture

The lab was built using virtual machines. Kali Linux was used as the Wazuh server, while Ubuntu Linux was configured as the monitored endpoint with the Wazuh agent. This separation allowed security events generated on the endpoint to be collected and investigated from the Wazuh environment.

Component	Role	Platform
Wazuh Server	Central security monitoring and analysis	Kali Linux
Wazuh Agent	Monitored endpoint	Ubuntu Linux
Suricata	Network security monitoring / IDS	Lab environment
VirusTotal	External threat intelligence enrichment	Integrated service

Wazuh Server Deployment

The Wazuh server was deployed on Kali Linux. The Wazuh components were installed and configured to provide centralized security monitoring. The Wazuh dashboard was used to review alerts, events, and endpoint activity.

- Wazuh Manager
- Wazuh monitoring and alerting
- Wazuh dashboard
- Wazuh agent communication

Endpoint Monitoring

Ubuntu Linux was configured as the monitored endpoint. The Wazuh agent enabled the server to receive security telemetry from the endpoint and provided visibility into security-related activity.

Security Detection and Testing

The laboratory included controlled security simulations to test detection and investigation capabilities. The activities were performed in the isolated lab rather than against production systems.

EICAR test file: Used as a safe, standardized antivirus test artifact to validate security detection workflows.

Diamorphine rootkit simulation: Used in the controlled lab to explore rootkit-related detection and alerting.

Suricata: Used to provide network-based security monitoring and detection.

VirusTotal integration: Used to enrich relevant file/hash information with external threat intelligence.

SOC Investigation Workflow

Stage	Activity
Monitor	Review Wazuh alerts and endpoint activity.
Triage	Identify alerts that require investigation and determine their relevance.
Validate	Examine the associated event, endpoint, file, process, or network activity.
Enrich	Use available threat intelligence, including VirusTotal integration, to add context.
Investigate	Determine the likely cause and security significance of the alert.
Document	Record the evidence, findings, and

	recommended response actions.
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Key Security Capabilities Demonstrated

- Endpoint security monitoring
- SIEM alert monitoring
- Security event investigation
- Malware detection workflow
- Rootkit detection concepts
- Network intrusion detection with Suricata
- Threat intelligence enrichment with VirusTotal
- Virtual lab deployment and troubleshooting
- SOC alert triage and analysis

Evidence and Screenshots

Supporting screenshots are stored in the project's GitHub screenshots folder. The evidence section should include the screenshots that demonstrate the actual configuration and detection work completed during the lab.

- Wazuh server/dashboard deployment.
- Ubuntu endpoint and Wazuh agent configuration.
- Wazuh alerts and security event monitoring.
- Suricata installation or detection evidence.
- EICAR test detection evidence.
- Rootkit simulation/detection evidence.
- VirusTotal integration or enrichment evidence.

Only screenshots that correspond to the actual lab work should be added to the final report.

Findings and Security Significance

The lab demonstrated how a SIEM can centralize endpoint security information and help a SOC analyst move from raw security events to actionable alerts. Combining endpoint monitoring with network detection and external threat intelligence provides broader visibility and can improve the investigation of suspicious activity.

Recommendations

- Maintain centralized monitoring of endpoints through Wazuh agents.
- Tune detection and alert rules to reduce false positives.
- Combine endpoint telemetry with network monitoring for better visibility.
- Use external threat intelligence to enrich suspicious files, hashes, and indicators.

- Document investigation findings and response actions for repeatable SOC processes.
- Continue expanding the lab with additional attack simulations and detection use cases.

Conclusion

The Wazuh SOC Lab provided practical experience deploying a SIEM in a virtual environment and investigating security events from a monitored Linux endpoint. The project combined Wazuh, Suricata, and VirusTotal to demonstrate endpoint monitoring, network detection, threat intelligence enrichment, and SOC investigation workflows. It forms a practical foundation for further work in SOC operations, incident response, and threat intelligence.